HEIQ XMEMBRANE PROJECT TEAM

Test-cell minimum area and ERV scale check

Short technical note · 1 September 2026

Decision. Retain the planned 50×50-mm maximum active window (25 cm^2). The 40×40-mm insert (16 cm^2) is the smallest practical aperture that covers the complete 0.5–5 A/m^2 low-current screen. A 30×30-mm insert (9 cm^2) is sufficient in the ideal signal calculation for the 1.24–3.37 A/m^2 range of the selected direct-active-ERV case.

1 Minimum measurable active area

The previous design already defined a 25-cm^2 maximum aperture and inserts down to 4 cm^2, but it did not state one explicit minimum area. The updated calculation holds the total flow at 6 L/min per side and requires a differential RH signal five times the combined expanded uncertainty of two 0.1-RH-percentage-point channels.

Case	j A/m^2	Ideal minimum cm^2	Insert cm^2
Inverse-design anchor, lower planned point	0.50	13.09	16
Inverse-design anchor	1.00	6.55	9
Selected direct-ERV profile minimum	1.242	5.27	9
Selected direct-ERV profile maximum	3.366	1.94	4
Order-unity reference	10	65.5	Does not fit at 6 L/min
Low-drag reference, $n = 0.3$	10	218	Does not fit at 6 L/min

The 25-cm^2 aperture can measure the weaker references by reducing flow: the ideal limits are 2.29 L/min for the order-unity case and 0.69 L/min for the low-drag case. Signal acquisition and high-flow film characterization must therefore be separate modes.

These are gross Faradaic signal floors. Passive back-diffusion, gas films, leakage, storage and electrode water can only reduce the net measured signal. The physical coupon should therefore remain approximately 80×80 mm to provide sealing and guard land, even when a smaller active insert is used.

2 Consistency with a full-scale ERV

The coupon and full-scale model are hydraulically consistent at the local channel level.

Quantity	Coupon	Direct ERV	Assessment
Channel gap	2 mm	2 mm	Direct match
Velocity	0.5–4 m/s	2 m/s	Full-scale point covered
Reynolds number at 2 m/s	472	488/529	Close laminar similarity
Flow length	50 mm	1.95/2.08 m	Coupon is a local test
Pressure drop	5.8 Pa	218/226 Pa	Consistent after length scaling

The full-drying case represents 150,000 CFM (70.79 m^3/s) and 34,425 m^2. RenewAire publishes membrane arrays up to 70,400 CFM, giving 2.13 arrays by airflow. Naples Community Hospital specified approximately 249 Pa maximum static pressure for its twelve-core enthalpy array, close to the model's 218–226 Pa. Kao Data London One uses thirteen plate-exchanger units at 13.5 m^3/s each; the project flow is 5.24 such units and 40% of that documented total.

Scale conclusion. Airflow, channel velocity, Reynolds number and pressure drop are credible for a very large modular building installation. The active membrane area remains an upper-end packaging challenge, not an obvious physical contradiction. It must be distributed across multiple cores, and commercial readiness cannot be claimed until electrodes, supports, manifolds, leakage, cleaning and manufactured pressure drop are included.

Sources and reproducibility. Commercial references: RenewAire SA Series specification; Naples Community Hospital case study; and FläktGroup Kao Data Centre case study.

The calculation is reproduced by `scripts/size_eo_vapor_cell_v1.py`; the complete table is in `outputs/eo_vapor_cell_sizing_v1/minimum_detectable_active_area.csv`.